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Biotechnology and
Biological Sciences
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Theory: Genetic maps and recombination

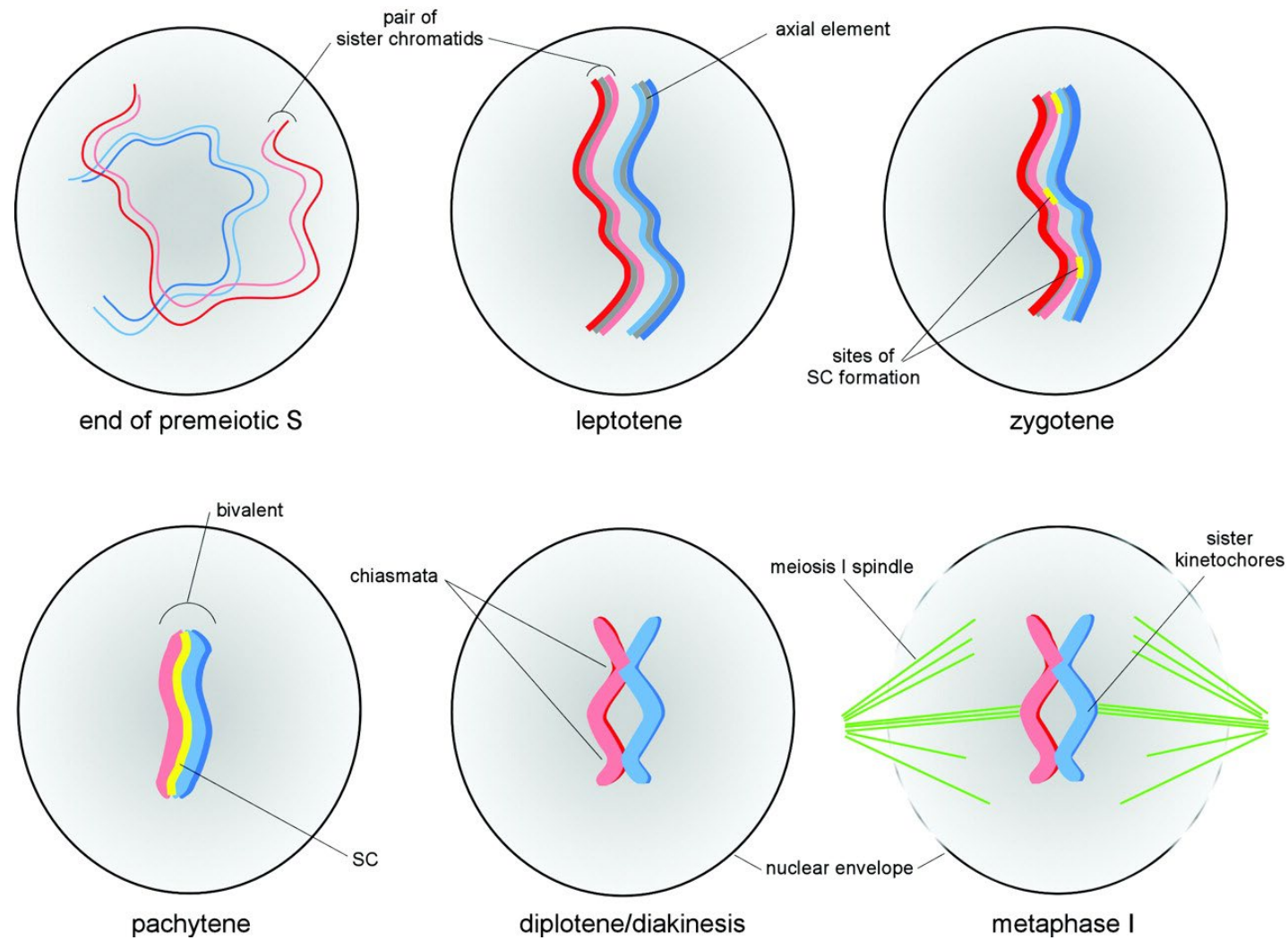
Chris Gaynor

Minneapolis, June 2026



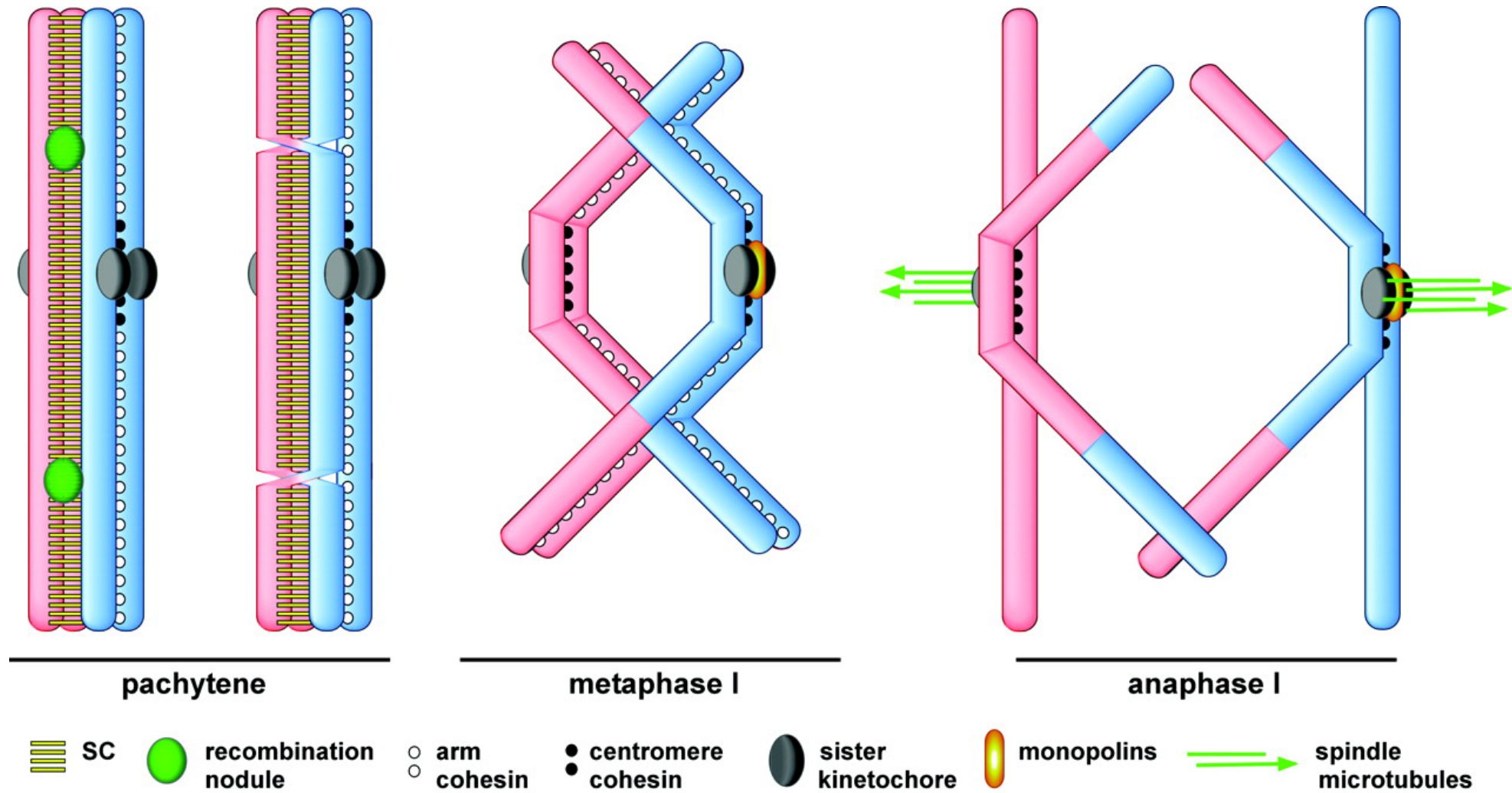
Bayer CropScience

Meiosis Review



Source: Page and Hawley (2003)

Meiosis Review



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Genetic Map Development

- Morgan observed “linked” traits working on fruit flies
 - Developed pearls on a string hypothesis for genes
 - Used recombination and crossing over as an explanation
- Sturtevant developed the first linkage map
 - A student in Morgan’s lab
 - Map used recombination rate as a mapping unit
 - Rate is not additive, and they acknowledged this
 - Also documented the occurrence of crossover interference

Genetic Mapping Functions

- Haldane developed the first genetic mapping function
 - Converts recombination rate to an additive mapping unit
 - Expected number of recombinations
 - One Morgan means one expected recombination
 - Doesn't account for crossover interference
- Kosambi first incorporated interference in a mapping function
 - Fits fly data very well
 - Different levels of interference in other species
 - Many subsequent mapping functions developed

Simple Modeling of Recombination

- Randomly choose a starting chromosome
- Determine if crossover occurs between 1st and 2nd locus
 - Convert genetic distance to recombination rate
- Repeat process for all intervals between loci
- Doesn't model crossover interference
 - Consistent with Haldane mapping function
- Easy to implement but very slow computation

Count-Location Model

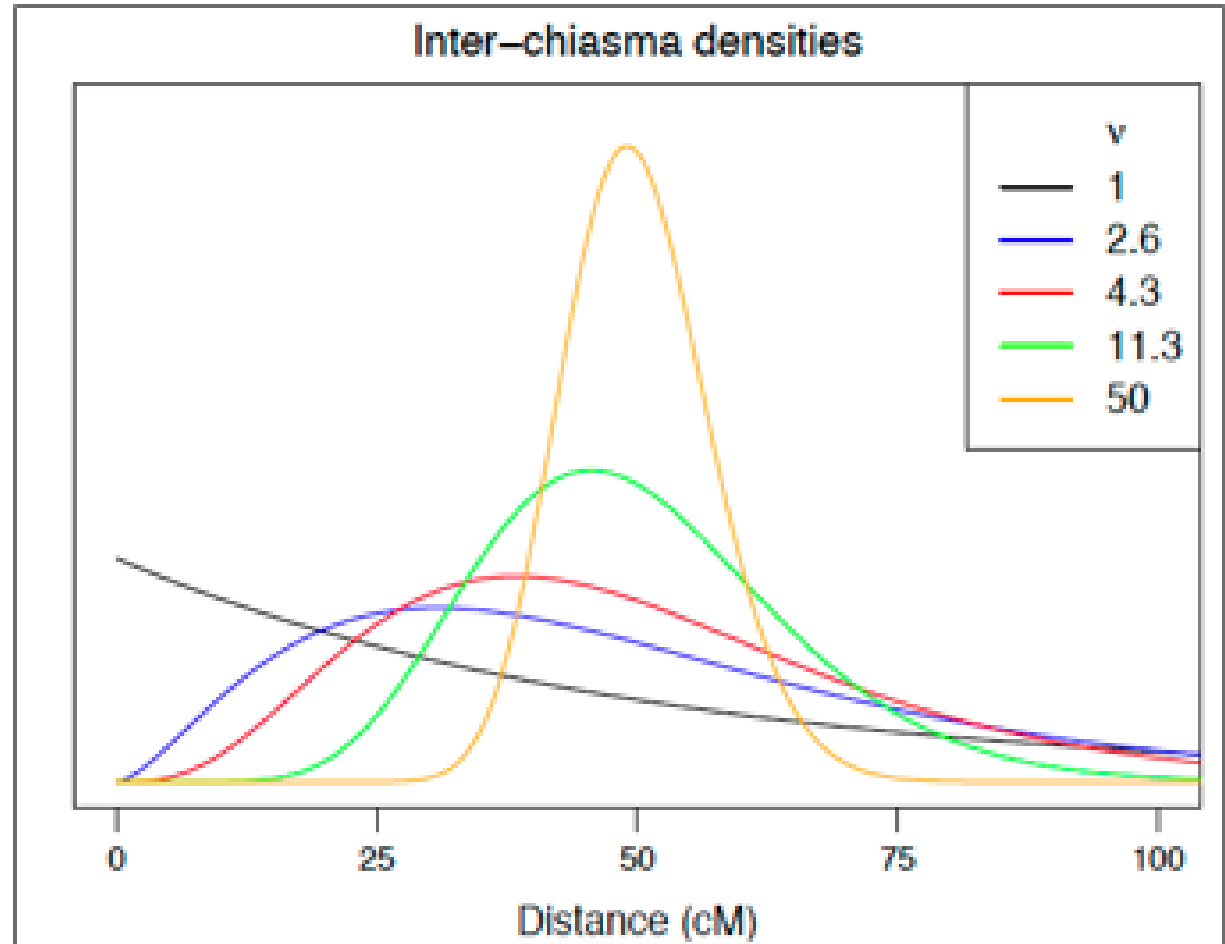
- More complex implementation but faster computation
 - Sample number of chiasmata from Poisson distribution (rate = $2M$)
 - Thin by half to account for the 4 chromatids (tetrad)
 - Sample positions of crossovers from uniform distribution (0- M)
 - Randomly pick a starting chromosome and resolve crossovers
- Does not model crossover interference
 - Consistent with Haldane mapping function
- Original AlphaSimR model

Gamma Model

- Positions of chiasmata sampled using a gamma distribution
 - Shape = v , rate = $2v$
- Incorporates crossover interference (v)
 - Able to approximate many mapping functions
- Can incorporate species specific crossover interference
 - The value v can be estimated from real data

Interference in Gamma Model

- Crossover interference parameter (v)
 - Haldane: $v = 1$
 - Kosambi: $v \approx 2.6$
 - Humans: $v = 4.3$
 - Mice: $v = 11.3$



Source: Karl Broman

Gamma “Sprinkling” Model

- Models two pathways for recombination
 - Type I, no crossover interference
 - Type II, crossover interference
- Involves sampling from two separate gamma distributions
 - Parameter controlling the proportion from each pathway
- Current AlphaSimR model
 - Default set to $v=2.6$ and no type I crossovers (~Kosambi)

AlphaSimR Demonstration